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Please, answer the theoretical questions in detail, using only the provided sheets. The exam is **open-book**, and you can use **no electronic devices** except for a portable calculator. You are encouraged to explain your reasoning, like in an oral exam. You are also encouraged to draw sketches for concepts that are visual in nature. Please refrain from copying from the notes! Use your own words and drawings.

The exam duration is **2hr**.

This written exam gives up to 32 points. The lab assignments provide up to extra 4 points. Total grades beyond 30/36 will be graded with 30 with laude.

Exercise 1

- A) Given the function $f(x, y) = (x - 2)^2 + (yx + 1)^3$, evaluate the gradient at $(x, y) = (1, 1)$.
- B) Describe the concept of stride in a convolution layer. Provide also graphical representation of the concept in the specific case of a stride of 4.
- C) Comment on the decrease in computational effort related to using strided convolutions. If you have a 64x64 input activation with 32 input and output channels and a 3x3 convolutional filter, compute the number of macs required to process the channel with stride 1 and with stride 2.
- D) Describe the concept of multiple heads in attention computation.

A)

$$df(x, y)/dx = d/dx ((x - 2)^2 + (yx + 1)^3) = 2(x - 2) + 3y(yx + 1)^2$$

$$df(x, y)/dy = d/dy ((x - 2)^2 + (yx + 1)^3) = 3x(yx + 1)^2$$

Therefore,

$$\frac{df(1,1)}{dx} = 2(1 - 2) + 3(1 + 1)^2 = -2 + 3 * 4 = 10$$

$$\frac{df(1,1)}{dy} = 3(1 + 1)^2 = 12$$

B) See theory

C) The stride parameter reduces the size of the output activation according to the following equation (see PyTorch reference at the following link <https://docs.pytorch.org/docs/stable/generated/torch.nn.Conv2d.html>):

$$H_{out} = \left\lfloor \frac{H_{in} + 2 \times \text{padding}[0] - \text{dilation}[0] \times (\text{kernel_size}[0] - 1) - 1}{\text{stride}[0]} + 1 \right\rfloor$$
$$W_{out} = \left\lfloor \frac{W_{in} + 2 \times \text{padding}[1] - \text{dilation}[1] \times (\text{kernel_size}[1] - 1) - 1}{\text{stride}[1]} + 1 \right\rfloor$$

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In case of 64x64 input, 32 input channels, 32 output channels and 3x3 filter and padding 1
(no decrease in image size due to kernel size):

STRIDE 1:

$$\begin{aligned} \text{MACs} &= H_{\text{out}} * W_{\text{out}} * C_{\text{in}} * C_{\text{out}} * K * K = H_{\text{out}}^2 * C_{\text{in}} * C_{\text{out}} * K^2 = \text{ceil}((64+2-1)/1+1)^2 \\ &* 32 * 32 * 3 * 3 = \text{ceil}(64)^2 * 32 * 32 * 3 * 3 = 64^2 * 32 * 32 * 3 * 3 = 64^2 * 32^2 * 9 = \\ &37,748,736 \end{aligned}$$

STRIDE 2:

$$\begin{aligned} \text{MACs} &= H_{\text{out}} * W_{\text{out}} * C_{\text{in}} * C_{\text{out}} * K * K = H_{\text{out}}^2 * C_{\text{in}} * C_{\text{out}} * K^2 = \text{ceil}((64+2-1)/2+1)^2 \\ &* 32 * 32 * 3 * 3 = \text{ceil}(63/2+1)^2 * 32 * 32 * 3 * 3 = \text{ceil}(32.5)^2 * 32 * 32 * 3 * 3 = 32^4 * 9 = \\ &9,437,184 \end{aligned}$$

IF NO PADDING IS CONSIDERED:

$$\text{STRIDE 1: } 9 * 32^2 * 62^2 = 35426304$$

$$\text{STRIDE 2: } 9 * 32^2 * 31^2 = 8856576$$

D) See theory

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Exercise 2

Consider an attention layer with $\text{seq}=32$ and $d=256$, followed by a feed forward layer (256×256 weights). In answering the questions explain your calculations.

- A) Compute the total number of MAC executed in the attention layer.
- B) Compute the total number of MAC executed in the feedforward layer.
- C) Compute the total storage needed to store the Q, K, V and **Wo input and weight** matrices assuming that activations are stored as FP32 and weights are stored as FP16.
- D) Compute the total number of exp function evaluations required.

A) Sequence of operations in the attention layer:

- Q, K, V projections $\rightarrow 3 * \text{seq} * d * d$ MACs = 6,291,456 MACs
- QK^T product $\rightarrow \text{seq} * d * \text{seq}$ MACs = 262,144 MACs
- $A * V$ product $\rightarrow \text{seq} * \text{seq} * d$ MACs = 262,144 MACs
- Softmax $\rightarrow$ No MACs considered (only adds in the denominator)
- W_o product $\rightarrow d * d * \text{seq}$ MACs = 2,097,152 MACs

Therefore, $\text{MACs_Att} = 8,912,896$

IF THEY SEPARATED THE OUTPUT PROJECTION IN THE FFN: $\text{MACs_Att} = 6,815,744$

B) The feedforward layer takes as input a $\text{seq} * d$ sized input, which is multiplied by its weights. As a result:

$\text{MACs_FF} = \text{seq} * d * d = 2,097,152$

C) Q, K, V:

Input acts: $4 * \text{seq} * d = 4 * 32 * 256$ B = 32,768 B

Weights: $3 * (2 * d * d) = 3 * 2 * 256 * 256$ B = 393,216 B

Output acts: $3 * (4 * \text{seq} * d) = 3 * 4 * 32 * 256$ B = 98,304 B

W_o :

Input act: $4 * \text{seq} * d = 4 * 32 * 256$ B = 32,768 B

Weights: $2 * d * d = 2 * 256 * 256$ B = 131,072 B

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Output act: $4 \text{ B} * \text{seq} * d = 4 * 32 * 256 \text{ B} = 32,768 \text{ B}$

Therefore, $\text{TOT_mem} = 720,896 \text{ B} = 721 \text{ kB}$

D) Exponential functions are evaluated only during the softmax calls. In particular, the attention layer computes the softmax on a matrix of size $\text{seq} * \text{seq} = 32 * 32$. The number of exponential functions per row is, then 32 for the denominator and 32 for the numerator. In total, rows are 32. Therefore:

$\text{NUM_exp} = (2^*)^{32} * 32 = (2048) 1024$

we can accept both inplace and not calculation of the softmax

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Exercise 3

A parallel cluster of RISC-V processors is executing a matrix multiplication of matrices A (16x256) and B (256x16). Assume that the computation is parallelized assigning rows of the output matrix C to processors. The cluster has 10 processors. Elements of A, B, C are 32bit FP numbers

III.a) Compute the L1 minimal shared memory size that is needed to store the complete matrices, and the total number of MACs needed to execute the matrix multiplication

III.b) Compute the parallel speedup achieved by the cluster.

III.c) Assume that you have to perform softmax normalization on the rows of matrix C and you parallelize the computation over rows of the matrix. Assume that calculation of $\exp(x)$ takes 50 cycles and neglect the cost of all other operations in softmax computation. How many cycles will it take to compute softmax normalization for the C matrix on the processor cluster?

III.a)

Tot memory = $4 \text{ B} * (16*256 + 256*16 + 16*16) = 4 \text{ B} * 8448 = 33,792 \text{ B}$

MACs = $16*256*16 = 65,536$

III.b)

Parallel speedup on 10 cores: since the rows to be parallelized are 16 and the cores 10, the first 6 cores will take 2 rows, while the last 4 will take only 1. Therefore, speedup is $16/2 = 8 \times$.

III.c)

Softmax operation: 50 cycles.

Softmax workload:

for i in 16:

 for j in 16:

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denominator(i) = sum(exp(x_ij))  
for j in 16:  
    softmax(i, j) = exp(x_ij) / denominator(i)
```

Latency to compute 1 row:

Latency_numerator + latency_denominator = 16 EXP + (16EXP) = 16 (32) EXP

Slowest cores (2 iterations), that is also the system latency -> 32(64) EXP = 32(64) * 50
cycles = 1600(3200) cycles